

Area-Constrained Design Space Exploration of an STT-RAM Memory

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Abstract—This report presents a quantitative design-space exploration study of a fixed-capacity STT-RAM memory under a hard area constraint. Using a 16 MB STT-RAM instance and NVSim as the main modeling tool, we first define a controlled nominal reference configuration and then perform sensitivity screening over physically meaningful device-level and organization-level parameter groups. The screening stage is used to identify which variables deserve detailed study before the final constrained sweep. Device-level continuous parameters are evaluated using average relative sensitivity, while discrete organization-level parameters are evaluated using average relative impact. These metrics are not intended for direct cross-category comparison; instead, they are used to select the leading candidate within each parameter pool. The screening results identify cell area as the strongest device-side candidate because it directly controls area feasibility while also producing secondary latency and energy effects. On the organization side, output-path muxing is the strongest grouped candidate, mainly due to its impact on dynamic energy. The final sweep jointly explores cell area, output-path muxing, and mat organization to study validity boundaries, area feasibility, and latency–energy tradeoffs under the hard area budget.

Index Terms—STT-RAM, NVSim, memory modeling, design space exploration, area constraint, sensitivity analysis

I. INTRODUCTION

STT-RAM is attractive for large on-chip memories because of its density advantage and negligible leakage, but memory quality depends on both device-level assumptions and organization-level choices. A favorable device model does not automatically produce a good macro once area, latency, and energy constraints are enforced.

This project studies that cross-layer interaction through a fixed-capacity 16 MB STT-RAM memory under a hard area budget. The main question is which device and organization choices define the feasible region and where the best latency–energy tradeoff appears.

II. PROJECT SCOPE AND MOTIVATION

The fixed-capacity setup keeps all design points at the same nominal storage target, so the comparison is not driven by capacity scaling. The hard area budget turns the problem into feasible-region analysis rather than an unconstrained search for a single best point.

This project is motivated by the cross-layer nature of STT-RAM evaluation: device assumptions affect density and intrinsic behavior, while organization choices affect peripheral overhead, wiring, and partitioning. We use NVSim to generate quantitative results, but interpret its outputs as model-based design-space evidence rather than as a complete physical design solution.

III. METHODOLOGY

We use a 16 MB STT-RAM memory macro modeled in NVSim [1]. The initial configuration is used as a nominal reference for local screening, not as a claimed global optimum. It provides a controlled operating point around which device-level and organization-level parameters can be perturbed and compared.

The reference configuration uses the NVSim RAM target with an STT-RAM memory-cell input file, which models a memory macro without cache-specific assumptions such as tags or associativity. Capacity is fixed at 16 MB, data width at 128 bits, process node at 22 nm, and the HP roadmap is used throughout.

The reference run uses ReadEDP as the nominal optimization target only to obtain a balanced, non-extreme starting point for local screening. It is not interpreted as the final objective of the project. Under this configuration, NVSim selects a valid reference organization with an 8×8 bank organization, a 2×2 mat organization, a 1024×512 subarray size, a sense-amplifier mux ratio of 16, and output mux ratios of 1 at both output levels. The resulting reference metrics are summarized as follows:

$$A_0 = 4.408 \text{ mm}^2,$$

$$L_{r0} = 4.523 \text{ ns}, \quad L_{w0} = 11.770 \text{ ns},$$

$$E_{r0} = 129.093 \text{ pJ}, \quad E_{w0} = 231.948 \text{ pJ}.$$

For compact comparison, we define average latency and average dynamic energy as

$$L_{\text{avg}} = \frac{L_r + L_w}{2}, \quad E_{\text{avg}} = \frac{E_r + E_w}{2}.$$

For the nominal reference point, this gives

$$L_{\text{avg},0} = 8.1465 \text{ ns}, \quad E_{\text{avg},0} = 180.5205 \text{ pJ}.$$

The hard area budget for the later constrained design-space sweep is defined as $1.10 \times$ the reference area:

$$A_{\text{budget}} = 1.10 A_0 = 4.8488 \text{ mm}^2.$$

Any design point whose total area exceeds this budget will be

treated as infeasible in the later constrained tradeoff analysis.

Before the final sweep, we screen candidate parameter groups so the detailed exploration is not chosen arbitrarily. Candidates are ranked separately within the device-level and organization-level pools using normalized response metrics.

For device-level screening, we use physically meaningful groups rather than treating every STT-RAM cell-file field independently. The candidate pool includes *CellArea*, *WriteCurrent* for reset/set current, *WritePulse* for reset/set pulse width, *ResistanceLevel* for on/off resistance, and *AccessWidth* for access-transistor width.

For paired device parameters such as reset/set current, reset/set pulse width, and on/off resistance, the same scale factor is applied to both fields. The resulting sensitivity is therefore interpreted as the response to a grouped device knob rather than to either individual field. For continuous device-level groups, we apply

$$S = \{0.8, 0.9, 1.1, 1.2\}$$

around the nominal value. The average relative sensitivity of metric m to parameter group g is defined as

$$S_{m,g} = \frac{1}{|S|} \sum_{s \in S} \left| \frac{m(s) - m_0}{m_0} \right| / (s - 1).$$

Here, m_0 is the metric value at the nominal reference point and $m(s)$ is the metric value measured after applying scale factor s . The numerator is the relative change in the metric, while the denominator is the relative change in the grouped device parameter. Therefore, $S_{m,g}$ measures the average metric response per unit relative change of the device-level parameter group. This normalization is necessary because different perturbation points use different magnitudes, such as 10% and 20% changes.

For discrete organization-level parameters such as mat organization, mux ratios, and routing style, a derivative-like percentage sensitivity is less meaningful. A one-percent change in a mat organization or mux hierarchy does not correspond to a clear physical perturbation. We therefore evaluate organization-level choices using average relative impact with respect to the same nominal reference configuration:

$$I_{m,p} = \frac{1}{N} \sum_i \left| \frac{m_i - m_0}{m_0} \right|.$$

Here, m_i is the metric value of a valid alternative configuration and N is the number of valid non-reference alternatives included in the average. This definition measures how much a discrete organization choice moves area, average latency, or average dynamic energy away from the nominal reference point.

The device-level sensitivity scores and organization-level impact scores are used only for within-pool comparisons. Device scores rank continuous device assumptions, while

organization scores rank discrete architectural choices; they are not intended to compare device and organization parameters directly.

For organization-level parameters, we also allow grouped organization knobs when multiple NVSim fields represent different levels of the same architectural structure. In particular, *ForceMuxOutputLev1* and *ForceMuxOutputLev2* both describe hierarchical output-path muxing after the sensing stage. We therefore define an *OutputMux* group for screening. During the one-factor screening stage, the two fields are swept separately, and the group-level impact is computed as the average of their metric-wise relative impacts:

$$I_{m,\text{OutputMux}} = \frac{1}{2} (I_{m,\text{MuxOutLev1}} + I_{m,\text{MuxOutLev2}}).$$

This avoids double-counting while allowing the screening score to represent the output-path muxing family as a single organization-level candidate. The final sweep then implements *OutputMux* as a co-swept grouped variable.

To compare candidates within each pool, we combine metric-wise responses into an aggregate screening score. For device-level continuous groups, we aggregate average relative sensitivities. For organization-level discrete groups, we aggregate average relative impacts. In both cases, the score is defined as

$$\text{Score}(g) = 0.4M_{A,g} + 0.3M_{L,g} + 0.3M_{E,g},$$

where $M_{A,g}$, $M_{L,g}$, and $M_{E,g}$ denote the area, average-latency, and average-energy terms for group g . For device-level groups, M corresponds to S ; for organization-level groups, M corresponds to I . We assign a slightly higher weight to area because the final design-space exploration is explicitly performed under a hard area constraint. The exact weights are not intended as a universal optimum; they are a practical screening choice that slightly prioritizes area while keeping latency and energy comparably important.

This screening score is not the only selection criterion. Final parameter selection also depends on whether the parameter can be swept cleanly in NVSim and whether the resulting trend is physically interpretable. The current screening results are used to narrow the candidate set and justify the final sweep.

After the screening stage, we perform a constrained design-space sweep using the leading device-side variable and the strongest organization-side variable, while also retaining a secondary organization-side knob that has a clean architectural interpretation. Specifically, the final sweep varies *CellArea*, a co-swept *OutputMux* group, and *MatOrganization*. For each sweep point, we fix capacity at 16 MB, assign values to the selected parameters, run NVSim, and extract area, latency, and energy metrics. Points that produce no valid NVSim solution are treated as invalid. Among the valid points, those whose total area exceeds the hard area budget are treated as area-infeasible. The remaining points define the feasible design region. Within this feasible region, we analyze latency–

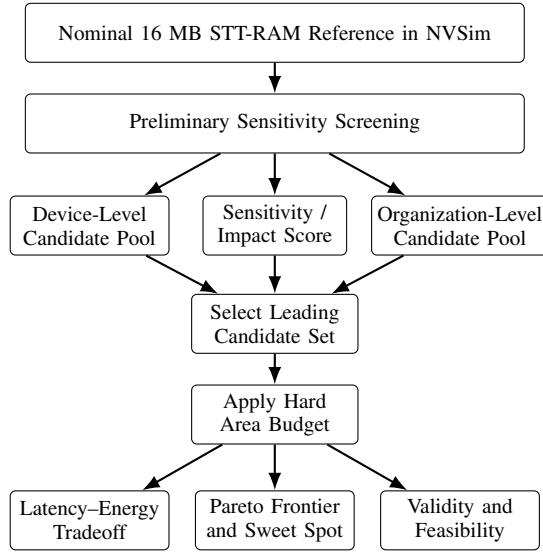


Fig. 1. Analysis flow. The screening stage uses a nominal reference configuration, grouped device-level and organization-level parameters, and normalized scoring before the final constrained sweep is fixed.

TABLE I
DEVICE-LEVEL AVERAGE RELATIVE SENSITIVITY SCORES FROM THE CURRENT NVSIM RUNS.

Device Parameter	Area	Latency	Energy	Score
CellArea	0.812	0.215	0.220	0.455
WritePulse	0.000	0.614	0.000	0.184
AccessWidth	0.000	0.025	0.011	0.011
WriteCurrent	0.000	0.004	0.001	0.002
ResistanceLevel	0.000	0.003	0.001	0.001

energy tradeoffs, identify Pareto-efficient designs, and select a representative balanced design point using a normalized latency-energy score.

IV. DEVICE-LEVEL SENSITIVITY RESULTS

We completed a device-level sensitivity sweep around the nominal reference point. The device screening dataset consists of one reference point and four non-reference perturbation points for each continuous device-level parameter group, using scale factors of 0.8, 0.9, 1.1, and 1.2. All five tested device-level parameter groups produced valid NVSim outputs.

For compact presentation in Table I and Fig. 2, *ResetCurrent/SetCurrent*, *ResetPulse/SetPulse*, *ResistanceOn/ResistanceOff*, and *AccessCMOSWidth* are reported as *WriteCurrent*, *WritePulse*, *ResistanceLevel*, and *AccessWidth*, respectively. Fig. 2 shows the area, latency, energy, and aggregate-score components in one grouped view.

Table I summarizes the device-level sensitivity scores. The score is computed from the average relative sensitivity over all non-reference scale factors. This makes the ranking less dependent on a single $\pm 10\%$ perturbation point and provides a more stable estimate of the local response around the nominal STT-RAM reference configuration.

Several observations follow from the device-level data. *CellArea* is the strongest device-side candidate, with an aggregate

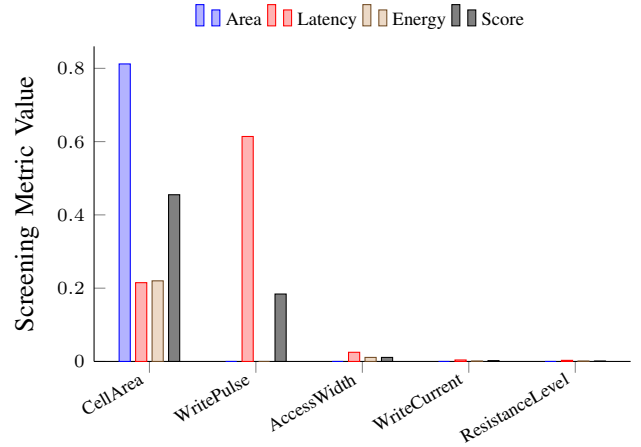


Fig. 2. Device-level screening results. Each parameter group is shown with area, latency, energy, and aggregate sensitivity-score bars.

score of 0.455. Its response is area-dominated, as expected, but the area sensitivity is 0.812 rather than 1.0, showing that peripheral structures also contribute significantly to total macro area. Scaling *CellArea* from $0.8\times$ to $1.2\times$ also changes average latency from 7.7955 ns to 8.4950 ns and average dynamic energy from 172.244 pJ to 188.1695 pJ, so it is not only an area knob but also a geometry-related assumption that affects delay and energy.

WritePulse is the second strongest device group, with an aggregate score of 0.184, and its response is almost entirely latency-driven: increasing the write-pulse scale from $0.8\times$ to $1.2\times$ changes write latency from 9.770 ns to 13.770 ns while read latency and area remain unchanged. *AccessWidth*, *WriteCurrent*, and *ResistanceLevel* produce much weaker macro-level responses in this NVSim setup. Therefore, *CellArea* is selected as the leading device-side variable for the final area-constrained sweep, while *WritePulse* remains a useful secondary observation for write-latency sensitivity.

V. ORGANIZATION-LEVEL SCREENING RESULTS

We also completed an organization-level screening sweep around the same nominal reference configuration. Unlike the device-level parameters, these organization-level choices are discrete architectural settings, so they are evaluated using average relative impact rather than average relative sensitivity. The current organization-level candidate pool includes: (1) *MuxSenseAmp*, which changes the sense-amplifier muxing ratio; (2) *OutputMux*, which groups the two output-path muxing levels *MuxOutLev1* and *MuxOutLev2*; (3) *MatOrganization*, which changes the mat-level partitioning; and (4) *Routing*, which changes the interconnect style.

For compact presentation, the grouped *OutputMux* score is reported as one organization-level candidate. It is computed by averaging the metric-wise impacts of the separate *MuxOutLev1* and *MuxOutLev2* one-factor sweeps. This is justified because both fields represent hierarchical output-path muxing after the sensing stage, and in the current reference configuration they produce identical macro-level responses for the tested values.

TABLE II
ORGANIZATION-LEVEL AVERAGE RELATIVE IMPACT SCORES FROM THE
CURRENT NVSIM RUNS.

Organization Parameter	Area	Latency	Energy	Score
OutputMux	0.092	0.011	0.244	0.113
MatOrganization	0.068	0.025	0.055	0.051
MuxSenseAmp	0.076	0.007	0.017	0.038
Routing	—	—	—	N/A

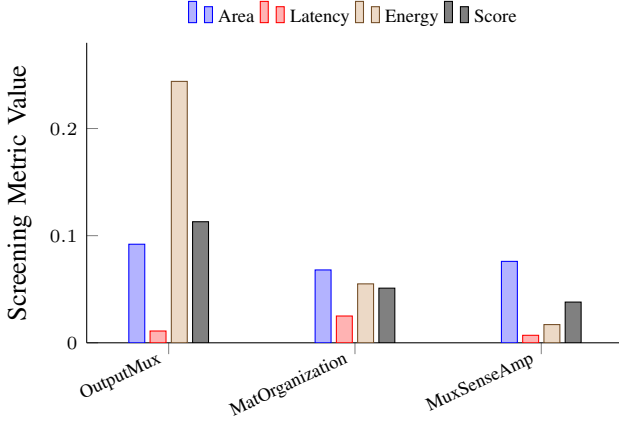


Fig. 3. Organization-level screening results. Each organization group is shown with area, latency, energy, and aggregate impact-score bars.

The grouped score is used to rank output-path muxing as an organization-level candidate, while the later final sweep can implement this candidate as a co-swept grouped variable.

Table II and Fig. 3 summarize the organization-level screening scores. *OutputMux* has the highest aggregate impact score, mainly because of its strong effect on average dynamic energy. *MatOrganization* has a lower score but affects all three metric categories in a physically interpretable way. *MuxSenseAmp* changes area but has weaker latency and energy impact in the current setup. *Routing* is not ranked because the non-H-tree alternative did not produce a complete valid solution in the current configuration.

Several observations follow from the organization-level data. *OutputMux* is the strongest organization group, with an aggregate score of 0.113 dominated by average dynamic-energy impact. Because *MuxOutLev1* and *MuxOutLev2* produce identical one-factor responses here, we interpret this as an output-path muxing family effect rather than as two independent conclusions.

MatOrganization is a strong secondary candidate. Its score is lower, but it affects area, average latency, and average dynamic energy in a more balanced and physically interpretable way. *MuxSenseAmp* has weaker latency and energy impact, and *Routing* is excluded from the ranking because the non-H-tree alternative did not produce a complete valid solution. Based on this screening, the final sweep co-sweeps *ForceMuxOutputLev1* and *ForceMuxOutputLev2* as *OutputMux*, while retaining *MatOrganization* to capture array-partitioning effects.

VI. FINAL CONSTRAINED DESIGN-SPACE EXPLORATION

After screening, the final constrained sweep uses three variables: *CellArea*, co-swept *OutputMux*, and *MatOrganization*. *OutputMux* is implemented as

$$ForceMuxOutputLev1 = ForceMuxOutputLev2 = k.$$

MatOrganization is retained because it provides a clean array-partitioning interpretation and can interact with output-path muxing.

Compared with the preliminary sweep, the final sweep extends the device-density range toward smaller STT-RAM cell-area assumptions while keeping the organization-level variables on simulator-supported architectural values. The final sweep uses

$$CellArea\ scale \in \{0.2, 0.3, \dots, 1.2, 1.3\},$$

$$OutputMux \in \{1, 2, 4, 8, 16\},$$

$$MatOrganization \in \{1 \times 1, 2 \times 2, 4 \times 4, 8 \times 8, 16 \times 16\}.$$

This gives a total of 300 attempted design points. Among them, 101 points produce complete valid NVSim solutions and 68 points are both valid and area-feasible under the hard area budget of

$$A_{budget} = 4.8488\text{ mm}^2.$$

The remaining points either exceed the area budget, fail to produce a valid NVSim solution, or produce incomplete outputs under the forced organization constraints. We treat those incomplete outputs as outside the usable design space for quantitative latency–energy comparison.

A. Validity, Area Feasibility, and Score Distribution

The final sweep reveals both validity boundaries and area-feasibility boundaries. All configurations with *MatOrganization*= 16×16 fail to produce complete valid solutions. In addition, aggressive co-swept *OutputMux* values of 8 and 16 generally produce invalid or incomplete outputs in the current setup. Therefore, the quantitatively usable region is concentrated around *OutputMux* $\in \{1, 2, 4\}$ and *MatOrganization* $\in \{1, 2, 4, 8\}$, with the exact feasibility boundary also depending on *CellArea*.

Figures 4–7 show the validity map, area-feasibility map, and normalized latency–energy score distribution for the usable mat-organization families. Area-feasible points are colored by $Score_{LE}$, while valid area-infeasible points and invalid or incomplete points are shown using separate markers. This view preserves the three design-space categories while also showing where the best balanced feasible regions appear.

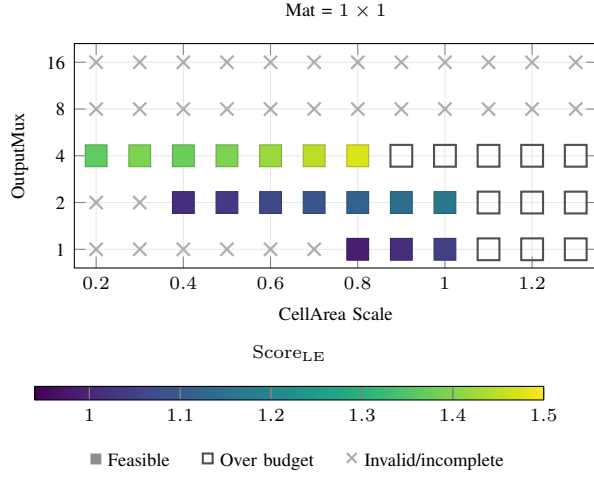


Fig. 4. Validity, area feasibility, and latency-energy score map for Mat = 1×1 . Feasible points are colored by their normalized score.

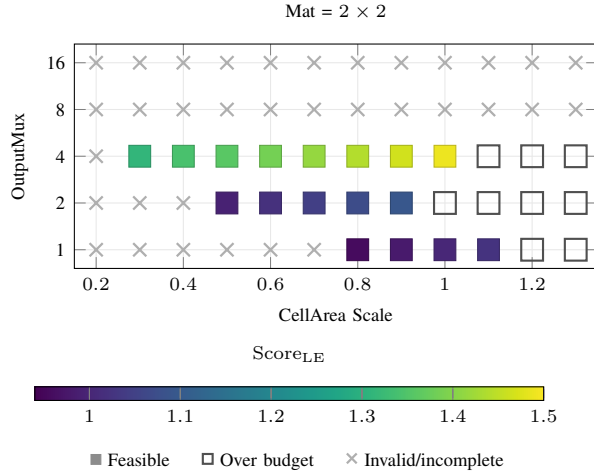


Fig. 5. Validity, area feasibility, and latency-energy score map for Mat = 2×2 . Feasible points are colored by their normalized score.

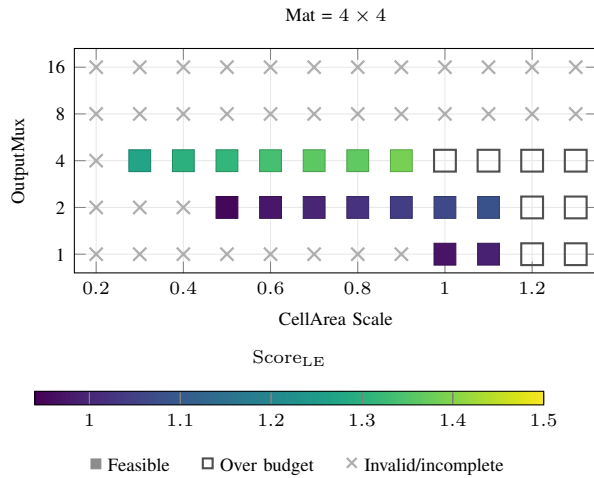


Fig. 6. Validity, area feasibility, and latency-energy score map for Mat = 4×4 . Feasible points are colored by their normalized score.

The maps show that *CellArea* controls the area-feasibility

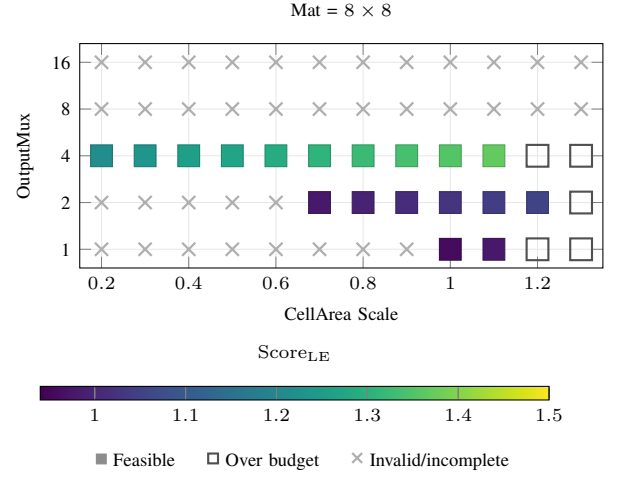


Fig. 7. Validity, area feasibility, and latency-energy score map for Mat = 8×8 . Feasible points are colored by their normalized score.

envelope, but the smallest cells are not automatically the best balanced choices. Very small cell area can reduce area and latency, but may increase dynamic energy depending on output muxing and mat organization. The maps also show strong validity boundaries: aggressive output muxing and the 16×16 mat organization do not yield complete valid solutions in this setup.

B. Latency-Energy Tradeoff

After filtering out invalid, incomplete, and area-infeasible points, the remaining 68 points define the feasible design region. We use average latency $L_{avg} = (L_r + L_w)/2$ and average dynamic energy $E_{avg} = (E_r + E_w)/2$ to compare feasible designs. The normalized latency-energy score used for balanced design selection is defined as

$$\text{Score}_{LE} = 0.5 \frac{L_{avg}}{L_{avg,0}} + 0.5 \frac{E_{avg}}{E_{avg,0}}. \quad (1)$$

The score in Eq. 1 is not intended to replace Pareto analysis. It is used only as a simple equal-weight ranking criterion among area-feasible points, after latency and energy have been normalized by the nominal reference values.

Figure 8 shows the latency-energy tradeoff among all area-feasible points. Pareto-efficient points are highlighted separately, the Pareto frontier is connected for clarity, and the best balanced point according to Eq. 1 is marked with a star.

The best balanced point is achieved by

$$\text{CellArea} = 0.5 \times, \text{OutputMux} = 2, \text{MatOrganization} = 4 \times 4.$$

This point has

$$A = 2.603 \text{ mm}^2, \quad L_{avg} = 7.236 \text{ ns}, \quad E_{avg} = 182.436 \text{ pJ},$$

and a normalized latency-energy score of 0.949. Compared with the nominal reference point, this design reduces average

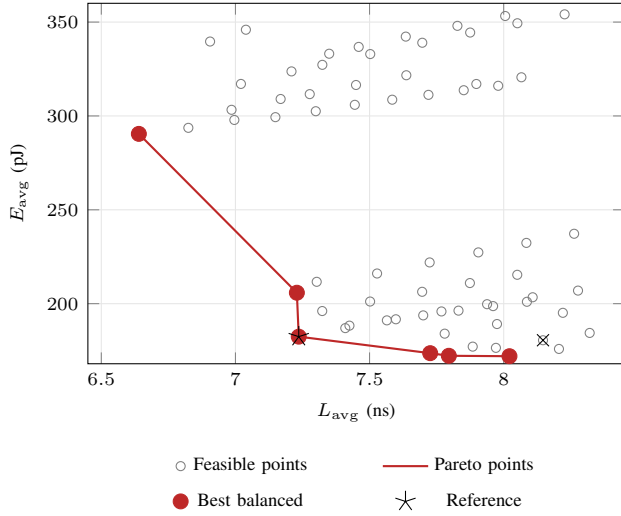


Fig. 8. Latency-energy tradeoff among area-feasible points. The Pareto frontier is connected for clarity, and the best balanced point is selected using the normalized latency-energy score.

latency from 8.1465 ns to 7.236 ns, an improvement of about 11.2%, while increasing average dynamic energy by only about 1.1%. It also uses only 2.603 mm², which is well below the 4.8488 mm² area budget. This result shows that extending the cell-area range changes the selected balanced design: the best point is no longer the nominal-cell-area organization, but a smaller-cell design with moderate output muxing and mat partitioning.

C. Selected Optimal Pareto and Balanced Designs

Table III lists the complete Pareto-efficient area-feasible designs. These points show the latency-energy boundary under the hard area constraint: the minimum-latency design uses the smallest tested cell area and aggressive output muxing, while the minimum-energy design remains near the nominal cell-area assumption with lower output muxing. The selected balanced design lies between these extremes.

TABLE III
COMPLETE PARETO-EFFICIENT AREA-FEASIBLE DESIGNS IDENTIFIED IN THE FINAL SWEEP.

Role	Design Parameters			Metrics		
	CellArea scale	OutMux	Mat	Area mm ²	L_{avg} ns	E_{avg} pJ
Min Latency	0.2×	4	8 × 8	1.314	6.640	290.451
Pareto Point	0.4×	2	1 × 1	2.687	7.229	205.831
Best Balanced	0.5×	2	4 × 4	2.603	7.236	182.436
Pareto Point	1.0×	1	8 × 8	4.137	7.726	173.634
Pareto Point	0.8×	1	2 × 2	3.693	7.796	172.244
Min Energy	1.0×	1	4 × 4	4.155	8.022	172.020

Overall, the final sweep shows that feasible design quality is shaped jointly by device density, output-path muxing, and mat-level partitioning. *CellArea* controls the area-feasibility envelope, but better density does not uniformly improve all metrics. *OutputMux* introduces strong validity and energy effects, while

MatOrganization shifts where the best feasible latency-energy points appear. The useful interpretation is therefore not that one knob should be pushed to an extreme, but that the device-density assumption changes which architectural organization is attractive under the same capacity and area constraint. This is also why the lowest-latency point is not chosen as the representative design: it lies on the Pareto boundary, but its energy cost is much higher than that of the balanced point.

The Pareto table also shows that several feasible points are close in either latency or energy but differ substantially in architecture. For example, the two low-energy points both use *OutputMux*=1, but they select different mat organizations. Conversely, the low-latency end of the frontier uses smaller cell-area assumptions and more aggressive valid output muxing. This spread supports the main conclusion that the final choice cannot be inferred from one parameter alone; it depends on how the selected device density changes the preferred memory organization.

VII. CONCLUSION

This report presents a staged, area-constrained design-space exploration of a 16 MB STT-RAM memory in NVSim. A nominal reference configuration is first defined, then device-level and organization-level candidates are screened using normalized response metrics, and finally the selected variables are swept under a hard area budget.

The screening results identify *CellArea* as the leading device-side variable because it strongly affects area feasibility while also producing measurable latency and energy effects. On the organization side, *OutputMux* has the largest aggregate impact, mainly through average dynamic energy, while *MatOrganization* is retained for its clear array-partitioning interpretation.

The final sweep jointly explores *CellArea*, co-swept *OutputMux*, and *MatOrganization*. Out of 300 attempted design points, 101 produce complete valid NVSim outputs and 68 are area-feasible. The sweep also reveals clear validity boundaries: 16 × 16 mat organization and aggressive co-swept output muxing values of 8 and 16 are largely outside the usable design space in this setup.

Among the feasible designs, the best balanced latency-energy point occurs at 0.5× cell area, *OutputMux*=2, and 4 × 4 mat organization. This design remains below the hard area budget while improving average latency relative to the nominal reference, at the cost of a moderate energy increase. Overall, the results show that design quality is determined by the interaction between device density and architectural organization rather than by a single independent knob.

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